

# Week 1: Tilings

Facilitator guide for the three adults

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## 1. The session at a glance

| Time      | What                                                                                                                                                                                                                     |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0:00–0:05 | Children arrive; free handling. Pattern blocks are on every table; the paper dominoes, L-pieces and X squares are already at the 2–3 and 4–5 tables; the K–1 table has blocks only (its dominoes come out at Problem 6). |
| 0:05–0:07 | Whole-group launch (section 3).                                                                                                                                                                                          |
| 0:07–0:52 | Table work. Each adult follows the order in sections 4–6 and keeps an eye on the clock marks given there.                                                                                                                |
| 0:52–1:00 | Share (section 7).                                                                                                                                                                                                       |

**Who sits where.** The parent takes the K–1 table: that table is about reading aloud, handing out pieces and writing numbers, and every answer and every “why” script is written out in section 4. The two mathematicians take grades 2–3 and grades 4–5, where the job is to push toward the arguments. (If the organizer prefers to sit with K–1, the parent can run 2–3 from section 5, which is equally complete.)

**How to run a table.** The child reads (or hears) the problem and does it. Give the hints in the order listed, one at a time, and only after the child has tried. Do not tell a child the reason behind a “no” before the problem that asks for it (the shaded problem at each table); the whole session is built so that the children meet the question first and the colouring second. Every problem ends with a short “compare” conversation (written out below) before the next problem. Numbers such as “4 covers” or “12 covers” are for you; the children are never asked to find them all unless a problem says so.

**Packets.** Print 4 copies of the K–1 pages, 5 of the 2–3 pages, 4 of the 4–5 pages (one spare each), and 3 copies of this guide.

## 2. Materials and preparation

**Pattern blocks.** During the launch every child needs 1 yellow hexagon, 3 blue rhombi, 2 red trapezoids and 6 green triangles (10 children: 10 Y, 30 B, 20 R, 60 G). After the launch the 2–3 and 4–5 tables put their blocks in a tray at the side (they are not needed again). The K–1 table keeps a tray with at least 30 greens, 20 blues, 12 reds and 4 yellows (per child during the problems: 9 G, 6 B, 3 R, 1 Y). The purple chevrons and the nonstandard pieces are not used this week; leave them in the box.

**Paper pieces,** cut from the 1-inch square grid paper (card stock if available; a Letter sheet of 1-inch grid gives about 30 dominoes or 20 L-pieces). Mark every L-piece with a coloured stripe so the two kinds are never mixed.

- Dominoes (1-by-2): K–1 table 24 (8 per child); 2–3 table 48 (12 per child); 4–5 table 30 (10 per child). Total 102, about 4 sheets. If time is short, the 2–3 and 4–5 children can cut their own dominoes in the free-handling minutes (cut 1-inch strips, then cut the strips in twos); the K–1 dominoes must be pre-cut.

- L-pieces (three squares in an L): K–1 table 12 (reserve, 4 per child); 2–3 table 32 (8 per child); 4–5 table 24 (8 per child). Total 68, about 4 sheets. Pre-cut these.
- X squares (a 1-inch square with a thick X in marker): 2 per child, 20 in all.
- Straight pieces (1-by-3, reserve for the 4–5 table only): cut 8 on demand.

**Other.** Crayons or coloured pencils (green, blue, red, yellow) at the K–1 table; pencils and erasers everywhere; two or three sheets of 1-inch grid paper on the 2–3 and 4–5 tables (for the predictions in 2–3 Problem 1 and for reserves); scissors on each table.

**Preparation time.** About 30 minutes of cutting for two adults, plus printing. Check one printed page against a block and a domino: the outlines and cells must be exactly 1 inch (print at 100%, “actual size”, not “fit to page”).

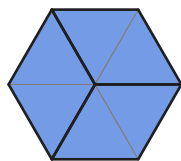
### 3. The whole-group launch (2 minutes)

Everyone has a yellow hexagon and a handful of blues, reds and greens in front of them. The organizer holds up a yellow hexagon.

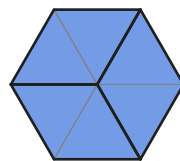
Script. “Everybody take one yellow hexagon and put it flat on the table. Now cover it exactly with blue rhombi: every bit of yellow hidden, no gaps, no overlaps, nothing sticking out over the edge. Go.” (About 20 seconds.) “How many blues? Three. Hold yours up. Look around: some of you made one picture and some made the other; there are exactly two ways to do it with blues.” “Now cover the yellow with reds only. How many? Two. Now greens only. How many? Six.”

“That is the rule for the whole hour: cover the shape exactly, then count. At your table you will get shapes and pieces, and the questions are always the same three. Can you cover it? If you can, how many ways are there? And if you cannot, how do you know for sure? Not ‘I tried a lot of times’. For sure. Go to your tables.”

That is the one concrete action everyone uses all hour: cover a printed shape exactly with pieces, then count or record. The two blue pictures on the hexagon (they are mirror images; each looks like a drawing of a cube) are the first “how many ways” and come back at the K–1 table.



blues, one way



blues, the other way

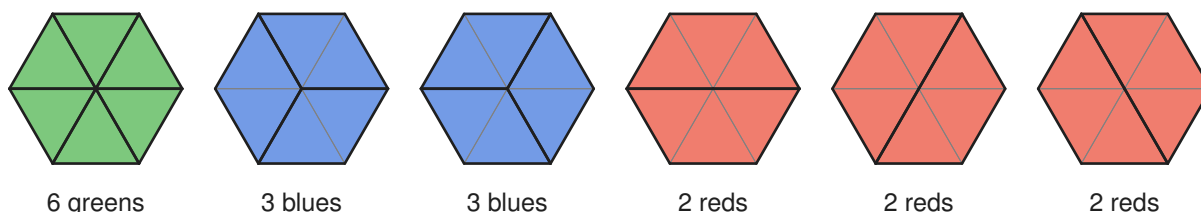
### 4. K–1 table: order, timing, answers, hints

Order: 1, 2, 3, 4, 5 (pattern blocks, about 31 minutes), then 6, 7, 8, 9 (dominoes, about 25 minutes). Table work runs from 0:07 to 0:52. Clock marks: Problem 3 by 0:16, Problem 5 by 0:30, Problem 6 by 0:37 at the latest, Problem 8 by 0:47. The two things this table must reach are Problem 5 (why blues fail on the triangles) and Problem 8 (why some crossed-out squares fail); to get there, trim in this order: stop Problem 2 after the 4-piece cover; leave the star in Problem 5 for the end; in Problem 7 pool the nine squares among the three children. Problem 9 is for a fast group.

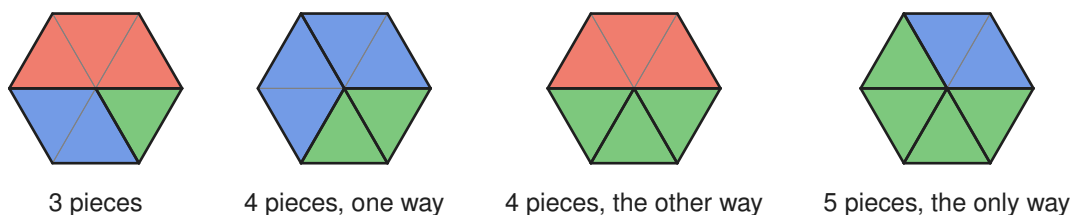
Working style: read each sentence aloud, wait, and let the children do the action; write the numbers yourself; the children colour and draw ticks and crosses. Three children can work on three copies at once; for Problem 7 it is better to pool: each child takes three of the nine squares and you keep one result grid in the middle of the table.

Names used on the pages: green triangle, blue rhombus (“rom-bus”; the children may say “diamond”), red trapezoid, yellow hexagon; domino; X square; result grid.

**Problem 1 — the hexagon in one colour.** Answers: 6 greens, 3 blues, 2 reds. There is one green cover, two blue covers (mirror images; a child who has one can be asked to find the other) and three red covers (the cut between the two reds can lie along any of the three long diagonals). Compare: “6, 3, 2. How many greens make one red? (3.) Check by building a red out of greens right on top of a red. How many greens make one blue? (2.) So a yellow is worth 6 greens, or 3 blues, or 2 reds.” Hints: if there is a gap or a piece over the edge, ask the child to point to it; if a child is unsure what “exactly” means, place one wrong piece yourself and ask what is wrong.

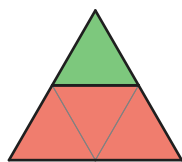


**Problem 2 — exactly 3, 4, 5 pieces.** Answers: Red + blue + green: yes (the red covers half the hexagon; the blue and green fill the other half; 12 placements, all the same picture up to turning and flipping). Exactly 4 pieces: two blues and two greens, or one red and three greens (both work). Exactly 5 pieces: one blue and four greens, and that is the only way. Argument in child form: “A red is worth 3 greens. If you use a red, the other 4 pieces have to share only 3 greens’ worth of room; that is impossible, every piece is at least one green. Two blues are worth 4 greens, so the other 3 pieces would share 2 greens’ worth; impossible too. So: one blue and four greens.” Compare: “1 piece works (the yellow itself), 2 pieces (reds), 3, 4, 5, 6 (all greens) all work. Could 7 pieces work?” Answer: no; six greens already fill it and green is the smallest piece. Hints, in order: (1) “How many greens is a red worth? A blue?” (2) “Start with the biggest piece you want, then fill the rest.” (3) For 5 pieces: “Try starting with a red; what goes wrong? Now start with a blue.”

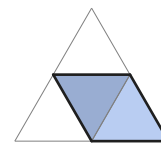
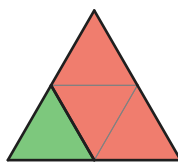
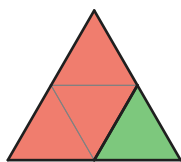


**Problem 3 — the side-2 triangle.** Answers: 4 greens. One red and one green: yes, three ways, the red lying along any one of the three sides and the green filling the opposite corner. Blues only: impossible. Argument in child form: “The big triangle is four small triangles: three corner triangles pointing up and one middle triangle pointing down. A blue rhombus is two small triangles that touch along an edge, and the corners do not touch each other, so every blue that fits covers the middle triangle plus one corner. Two blues would both need the middle triangle.” Note for you: the area is even ( $4 = 2 + 2$ ), so counting area is not enough; this is the first “no” that needs a real reason, and Problem 5 gives the general form. Hints: (1) “Put one blue anywhere. Which small triangles is it lying on?” (2) “Now try to fit the second blue. What is

in the way?" (3) "Is there any place at all for a blue that does not use the middle triangle?"

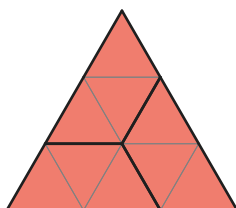


one red and one green

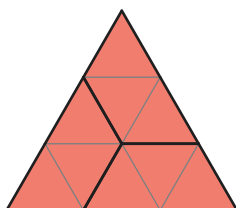


every blue uses the middle

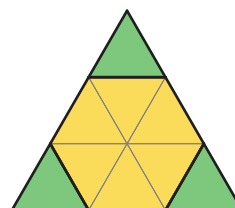
**Problem 4 — the side-3 triangle.** Answers: 9 greens. Reds only: 3 reds, and there are exactly two ways, mirror images of each other. One way: lay one red flat along the base, pushed into the left corner (it covers the left three of the five bottom units); the two remaining bottom units on the right together with the right unit of the middle row make a tilted red along the right side; the top unit with the left two units of the middle row makes a tilted red along the left side. The other way is the mirror image (first red pushed into the right corner). Yellow inside, rest green: exactly one way; the yellow only fits in the middle (it needs a point with six small triangles around it, and the side-3 triangle has only one such point inside), and then the three corners are left, one green each. Blues only: impossible; 9 small triangles, and blues cover 2 at a time: 2, 4, 6, 8, so at least one triangle is always left over (a child will get 3 blues and three leftover triangles; say "count by twos with me"). Compare: "Two different 'no' reasons so far: on the small triangle (Problem 3) there was room for two blues but the middle triangle was in the way; here there is an odd number of triangles." Extension for a fast child: blues plus exactly one green: still impossible (8 triangles left, an even number, and still no); the reason is Problem 5's. Hints for reds: (1) "Lay one red flat along the bottom, pushed into a corner." (2) "What is left is a bent strip; one red fits tilted along each side." Hint for the yellow: "Try to fit the yellow in a corner. Does it fit? Where is the only place?"



3 reds, one way



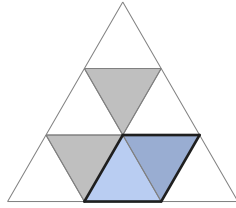
3 reds, the mirror image



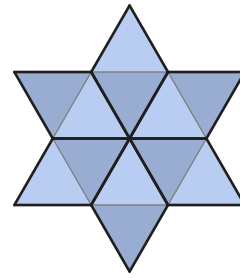
the yellow inside, 3 greens

**Problem 5 — the shaded shapes (the reason).** Answers: a blue always covers one white and one grey triangle: two small triangles that share an edge always point opposite ways, one up (white) and one down (grey). Counts: hexagon 3 white, 3 grey; side-2 triangle 3 white, 1 grey; side-3 triangle 6 white, 3 grey; star 6 white, 6 grey. Blues on the star: yes, 6 blues, and there is only one way: each point of the star touches only one other triangle (the one just inside it), so the blue on each point is forced. Compare, with the four shapes in a row: "Blues worked on the hexagon (3 and 3) and on the star (6 and 6). Blues failed on both triangles (3 and 1; 6 and 3). Every blue eats one white and one grey. So blues can only ever cover a shape with the same number of white and grey." For the side-3 triangle: "After 3 blues the greys are used up and 3 whites are left, and no blue can lie on two whites." This is the same reason as the middle triangle in Problem 3 (the middle triangle is the only grey one). Hints: (1) "Say the colours out loud each time you move the blue: white, grey; white, grey." (2) For counting: "Touch each white triangle as you count; then the greys." (3) For the star: "Start at a point of the star. Where can the blue that covers

that point go?”



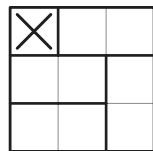
a blue on the shaded triangle: one white, one grey



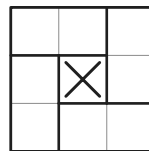
the star: 6 blues, the only way

**Problem 6 — first dominoes.** Answers: 2-by-3: 3 dominoes (3 ways). 2-by-4: 4 dominoes (5 ways). 3-by-3: cannot be covered; with 4 dominoes one square is left over; a child who gets stuck with 3 dominoes and 3 empty squares should rearrange so that a fourth fits, and there is still one left. Argument: “Count the squares: 6, 8, 9. Dominoes cover squares two at a time, 2, 4, 6, 8, so 9 squares can never all be covered.” Hints: (1) “Count the squares on each board.” (2) “Let us count by twos as we place dominoes.”

**Problem 7 — 3-by-3 with one square crossed out.** Before reading the problem, hand out the X squares: “This paper square with an X is a square we are not allowed to cover. Wherever it sits, that square is out.” Answers: ticks at the four corners and the centre (5 squares); crosses at the four middle-of-a-side squares. Example covers are pictured below. Number of covers, for you: corner out 4, centre out 2 (the ring of eight squares can be covered “clockwise” or “anticlockwise”). Compare: “Look at the ticks: corners and centre. Look at the crosses: the middles of the sides. It looks like a checkerboard.” Do not explain yet; Problem 8 does. Hints: (1) for a middle-of-a-side X, after two failed tries: “Which square got trapped?” (with (1,2) crossed out, the two top corners each have only one free neighbour, straight down, and after those two dominoes the bottom row and the centre cannot be paired up). (2) To keep the pace, pool the work: each child tries three positions.



corner (1,1) crossed out

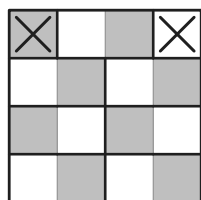


centre crossed out

**Problem 8 — the checkerboard (the reason).** Answers: a domino always covers one dark and one light square, because squares that share an edge have different colours. Board A (corner out): 4 dark, 4 light. Board B (middle of a side out): 5 dark, 3 light. Board C (centre out): 4 dark, 4 light. Compare with Problem 7: “The ticks are exactly the dark squares. Four dominoes eat 4 dark and 4 light. Board B has 5 dark and 3 light, so after 3 dominoes the light squares are gone, 2 dark squares are left, and dark squares never touch each other, so no domino can cover them.” Then: “This is the same trick as the blue on the triangle: one white and one grey every time.” Hints: (1) “Say the colours as you move the domino: dark, light.” (2) “Count the dark squares first, touching each one.”

**Problem 9 — two corners of the 4-by-4.** Answers: picture A (opposite corners): impossible; 6 dark, 8 light are left (both crossed-out corners are dark). Picture B (two top corners): possible; 7 dark, 7 light; an example is pictured below (one domino in the middle of the top row, then two flat dominoes in each other row). Compare: “Two squares gone each time; on A both were dark, on B one dark and one light. The colours decide, not the corners.” Hints: (1) for A, after a real try: “Count the colours first.” (2) For B: “Start

with the two middle squares of the top row.”

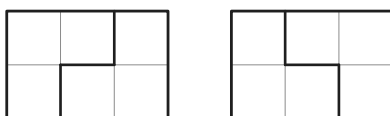


picture B: a cover

**If the table stalls.** A child who cannot find a cover that exists: take the pieces off and ask the child to start with a piece in a corner. A child who insists an impossible cover “almost worked”: ask them to point to the leftover square or triangle and ask “could any piece ever cover just that?” A child who is done with colouring before the others: ask for a different cover of the same shape (hexagon with blues, one red and one green on the side-2 triangle, 2-by-4 with dominoes).

**If the table finishes early (reserves, in this order).**

1. L-pieces (use the 2-by-3 and 3-by-3 boards from Problem 6; 4 L-pieces per child). “Cover the 2-by-3 board with two L-pieces” (2 ways, pictured below). “Now the 3-by-3 with three L-pieces”: impossible; an L-piece can never cover two corners of the board (try it), so three L-pieces cover at most three corners and there are four.



2. Build-and-copy with yellows (no outlines needed). “Put two yellows together, edge to edge. Build the same shape next to it out of reds only. How many? (4.) Blues only? (6.) Greens only? (12.)” Then three yellows in a row: 6 reds, 9 blues, 18 greens. Then the flower: “Put one yellow in the middle and six reds around it, each red’s short side against the yellow, to make a big hexagon. Build the same big hexagon with reds only (8 reds), then blues only (12 blues), then greens (24).” All verified; the big hexagon cannot be made of yellows only (at most 3 fit and gaps remain).
3. The domino game, you against a child, on the 3-by-3 board: take turns placing one domino anywhere it fits; whoever cannot place a domino loses. Let the child go first; the second player can always win on 3-by-3. Winning replies for you: if the child’s first domino lies along a side (a corner and the middle of that side), answer with the domino that goes from the centre straight away from that side; if the child’s first domino goes from the middle of a side into the centre, answer with any domino lying along the opposite side. After that the free squares fall into separate small parts; whenever the child plays in one part, you play in another, and the child runs out of moves first. On the 2-by-3 board the first player wins by placing an upright domino.

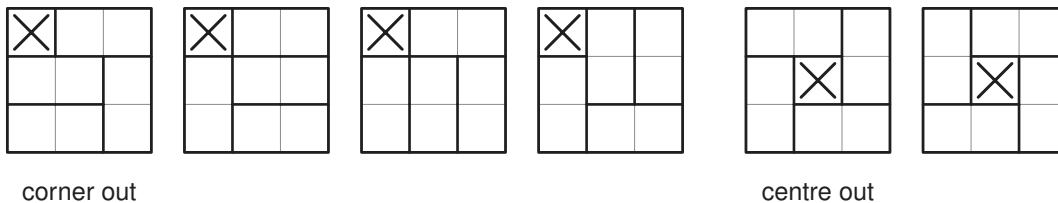
## 5. Grades 2–3 table: order, timing, answers, hints

Order: 1, 2, 3, 4, 5 (dominoes, about 39 minutes), then 6, 7, 8 (L-pieces, about 28 minutes). Table work runs from 0:07 to 0:52. Clock marks: Problem 3 by 0:25, Problem 5 by 0:37, Problem 6 by 0:46 at the latest (if Problem 5 is still running at 0:45, stop after four tests and go to Problem 6; the L-pieces are worth reaching). Problem 8 and the reserves are for a fast group.

**Problem 1 — which boards can dominoes cover.** Answers: 2-by-3: yes, 3 dominoes (3 covers).

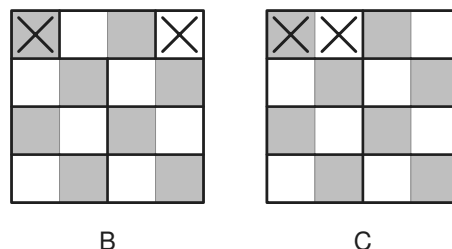
3-by-3: no, 9 squares, one square left over (with 4 dominoes; a child may also get stuck at 3 dominoes with 3 squares left; either way it fails). 3-by-4: yes, 6 dominoes (11 covers). Predictions: 3-by-5: no (15 squares); 4-by-4: yes (16 squares, 8 dominoes; 36 covers); 5-by-5: no (25). Argument: every domino covers 2 squares, so the number of covered squares is always even; a board with an odd number of squares always has a square left. Ask: “How do you know for sure that no clever arrangement covers the 3-by-3?” The answer “9 is odd; dominoes count by twos” is a complete proof, and say so. Hints: (1) “Count the squares of each board first.” (2) “Which numbers of squares can dominoes cover: 2, 4, 6, ...?”

**Problem 2 — 3-by-3 with one square crossed out.** Answers: ticks at the four corners and the centre; crosses at the four middle-of-a-side squares. All four covers with corner (1,1) out, and both covers with the centre out, are pictured below. Compare: “Where are the ticks? Corners and centre. The crosses? Middles of the sides. Does the tick pattern remind you of anything?” (a checkerboard). Do not explain yet. Hints: (1) after a failed try on a middle square: “Try a different first domino” (twice). (2) “Are you sure? Say which square gets trapped”: with (1,2) out, each top corner has only one free neighbour, straight below it; place those two dominoes and the centre plus the bottom row cannot be paired. (3) If the nine trials drag, split them: each child takes two or three squares and the table pools the results on one grid.



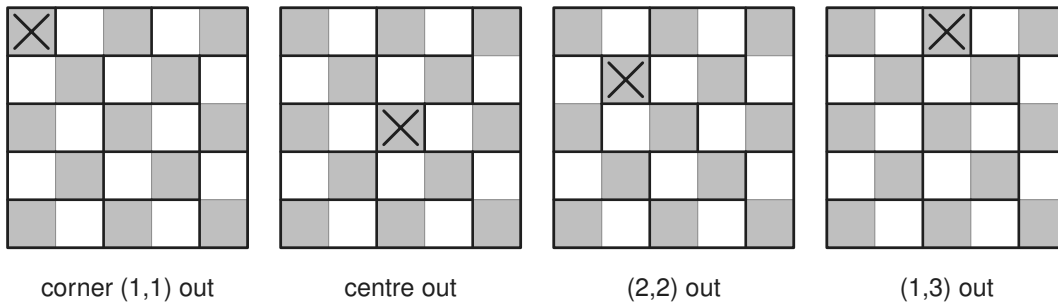
**Problem 3 — the checkerboard (the reason).** Answers: a domino always covers one dark and one light square, because neighbouring squares have different colours. A (corner out): 4 dark, 4 light, covered. B (middle of a side out): 5 dark, 3 light, not covered. C (centre out): 4 dark, 4 light, covered. Script: “Every domino uses up one dark and one light square. Four dominoes use 4 dark and 4 light. Board B has 5 dark and 3 light, so it can never be covered. That is a proof: it works for every arrangement anybody could ever try.” Then: “Look back at your result grid. The ticks are exactly the dark squares of the checkerboard.” Ask: “Would the same idea tell us which squares can be crossed out on a 5-by-5?” (that is Problem 5). Hints: (1) “Say the colours as you move the domino.” (2) “Count the dark squares by touching them.”

**Problem 4 — two corners of the 4-by-4.** Answers: A (opposite corners): impossible; 6 dark, 8 light remain (both crossed-out corners are dark). B (two top corners): possible; 7 and 7; 12 covers, one pictured below. C (a corner and its neighbour): possible; 7 and 7; 18 covers, one pictured below. Compare: “Two squares are missing each time. What is different about A?” (both missing squares are dark). “So crossing out two squares is fine when they have different colours, and hopeless when they have the same colour.” Hints: (1) “Count the colours before you start.” (2) For B and C: “Fill row by row with flat dominoes.”

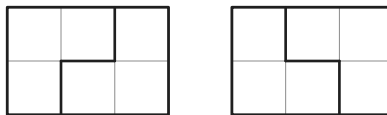


**Problem 5 — 5-by-5, predict then test.** Answers: exactly the 13 dark squares work ( $r + c$  even: the four corners, the middle square of each side, the four squares diagonally next to the centre, and the centre);

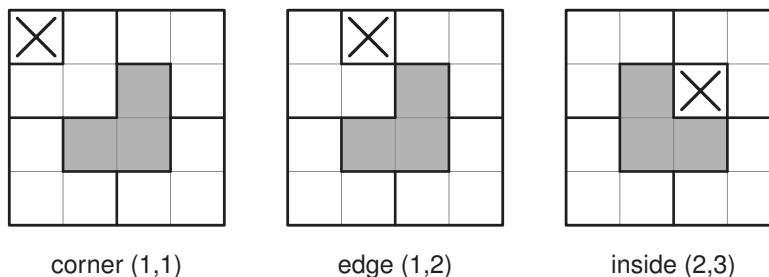
the 12 light squares fail. Numbers of covers, for you: a corner 192, a side-middle such as (1,3) 192, a square such as (2,2) 112, the centre 196. Example covers are pictured below. Argument: the board has 13 dark and 12 light squares; 12 dominoes need 12 of each; so the crossed-out square must be dark. That the dark squares all really work is what the tests show (the reason only says “must be dark”; the building says “dark is enough here”). Compare: put the prediction grid and the result grid side by side. A prediction of “only corners and centre” (copied from the 3-by-3) is the useful mistake: the tests on (1,3) or (2,2) correct it. Hints: (1) “Shade the 5-by-5 like a checkerboard in pencil.” (2) “How many dark and how many light squares are there?” (3) For a cover: “Fill row by row with flat dominoes; a row with an odd number of free squares lends one square to the next row through an upright domino.”



**Problem 6 — L-pieces on 2-by-3 and 3-by-3.** Answers: 2-by-3: exactly two covers, mirror images (pictured below). 3-by-3: impossible. Argument: an L-piece always fits inside some 2-by-2 block; two corners of the 3-by-3 never lie in the same 2-by-2 block; so one L-piece covers at most one corner; three L-pieces cover at most three corners; the board has four. (Nine squares, three pieces of three: the area is fine; the shape is what fails.) Hints: (1) “Try to cover two corners of the board with one L-piece. Can you?” (2) “So how many corners can three L-pieces cover at most?”

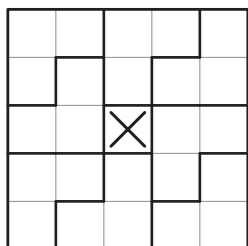


**Problem 7 — 4-by-4 with one square crossed out, five L-pieces.** Answers: all three are possible and each has exactly one cover (true for every one of the 16 squares); the covers are pictured below with the middle L-piece shaded. In every cover one L-piece sits in the middle four squares, covering one square of each of the three 2-by-2 quarters that do not contain the hole; then each quarter is a 2-by-2 with one square missing, filled by one L-piece. The surprise is that the “second way” never exists. Compare: “Look at your three drawings. What do they have in common?” (the L-piece in the middle). Hints: (1) “The 2-by-2 quarter that contains the hole needs one L-piece that fills its other three squares. Start there.” (2) “Now look at the middle four squares of the board.” (3) For the second-way search, let them try for two minutes, then: “Where could the middle L-piece be instead? Try each; something always gets stuck.”

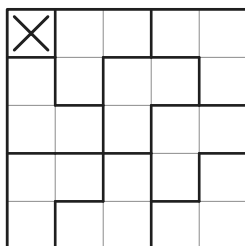


**Problem 8 — 5-by-5 with one square crossed out, eight L-pieces.** Answers: centre: yes (32 covers).

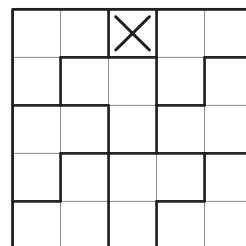
Corner: yes (8 covers). Middle of the top edge (1,3): yes (16 covers). Next to the corner along the top edge (1,2): no. Example covers are pictured below. In general exactly the nine squares whose row and column are both odd work; the other sixteen fail. Argument, for you and for a child who asks: put a dot on every square with odd row and odd column (nine dots); every L-piece fits inside a 2-by-2 block, and every 2-by-2 block contains exactly one dot, so an L-piece covers at most one dot; eight L-pieces cover at most eight dots; the ninth dot must be the crossed-out square. Compare: “On the 4-by-4 every square worked. On the 5-by-5, some do and some do not; which ones?” Hints: (1) for the centre: “Cover the four corners first, one L-piece in each corner.” (2) For (1,2), after two tries: “Draw a dot on every square where the row number and the column number are both odd. How many dots can one L-piece cover?”



centre out



corner out



(1,3) out

**If the table stalls.** On a “yes” board, remove everything and start with a corner; on a “no” board, ask the child to name the trapped square, and if there is none (Problem 4A, Problem 5 light squares), say “then we need a reason that is not about one square” and go to the colour count. Children who race ahead in Problem 2 or 5 should draw their covers on the small boards; the drawings are what the table compares.

**If the table finishes early (reserves, in this order).**

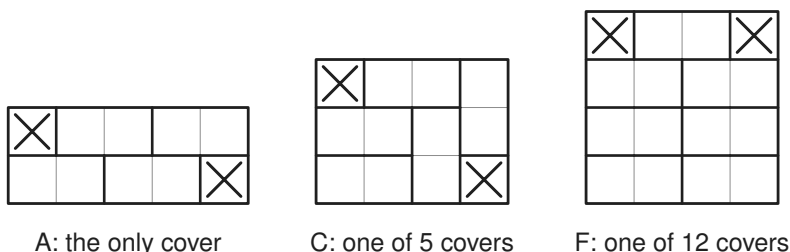
1. The centre trick (Problem 7’s board, X square, 5 L-pieces): “Put the X anywhere. Place one L-piece in the middle covering one square in each of the three quarters without the X. Now finish.” It always works, and it is the only cover. Ask: “Would this trick work on an 8-by-8 board, quarters inside quarters?” (Yes; 21 L-pieces; the 4–5 table is doing it.)
2. All the covers of 2-by-2, 2-by-3, 2-by-4 with dominoes (draw them on the grid paper): 2, 3, 5 covers. Predict 2-by-5: 8 (then 13, 21). Rule in words: the left end is either one upright domino (leaving a board one column shorter) or two flat dominoes on top of each other (leaving a board two columns shorter), so each number is the sum of the two before it.
3. The domino game on the boards of Problem 1: two players take turns placing one domino anywhere it fits; whoever cannot move loses. 2-by-3: the first player wins by placing an upright domino. 3-by-3: the second player wins (if the first domino lies along a side, reply with the domino that goes from the centre straight away from that side; if the first domino runs from the middle of a side into the centre, reply with any domino lying along the opposite side; after that, answer each move in a different part of the board and you have the last move). 3-by-4: the first player wins by covering the two central squares and then copying every move of the opponent rotated by a half turn about the centre. 2-by-4 (and 4-by-4): the second player wins by the half-turn copying.
4. Most L-pieces that fit without overlapping: 3-by-3: 2 (3 squares empty); 3-by-4: 4 (a full cover); 4-by-4: 5 (one square empty); 5-by-5: 8 (one square empty, and it is always one of the nine dotted squares).

## 6. Grades 4–5 table: order, timing, answers, hints

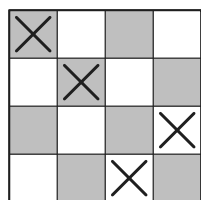
Order: 1, 2, 3, 4 (dominoes, about 27 minutes), then 5, 6, 7, 8, 9 (L-pieces, about 39 minutes). Table work runs from 0:07 to 0:52. Clock marks: Problem 2 by 0:18, Problem 5 by 0:34 at the latest, Problem 7 by

0:45, Problem 8 by 0:50 if it is to be done at the table. The two things this table must reach are Problem 2 (the colour reason) and Problem 7 (the centre trick). To get there: if behind at 0:30, skip Problem 4 (the organizer shows the 8-by-8 at the share) and do Problem 3 with G and H only; if behind at 0:40, do Problem 6 with pictures A and C only. Problem 9 and the reserves are for a fast group; a group that reaches Problem 8 only at 0:50 should draw the first three L-pieces (the middle one and the middle ones of two quarters) and explain at the share why the rest must work.

**Problem 1 — two corners crossed out, six boards.** Answers: A (2-by-5, opposite corners): yes, and there is only one cover, a staircase forced from the corners (pictured below). B (2-by-5, both top corners): no; the square (2,1) has only one free neighbour, so (2,1)–(2,2) is forced, then (1,2)–(1,3) is forced, then (1,4)–(2,4), and (2,3) is trapped with no free neighbour. C (3-by-4, opposite corners): yes, 5 covers, one pictured below. D (3-by-4, both left corners): no; forced chain (2,1)–(2,2), (1,2)–(1,3), (1,4)–(2,4), (2,3)–(3,3), and then (3,2) and (3,4) are trapped. E (4-by-4, opposite corners): no, and this time nothing is trapped at the start: every free square has at least two free neighbours, and every attempt simply ends with leftover squares that do not touch each other. F (4-by-4, both top corners): yes, 12 covers, one pictured below. Compare, with the six small boards in a row: “Opposite corners: yes, yes, no. Same side: no, no, yes. So it is not about where the corners are. On B and D you could point at a trapped square. Can you on E?” (No.) “Then we need a different kind of reason.” That is Problem 2. Hints: (1) for A: “Start at the bottom-left square; it has only one neighbour.” (2) For B and D: “Find a square with only one free neighbour; that domino is forced. Keep going.” (3) For E, let them try for three minutes, then hold the compare conversation.



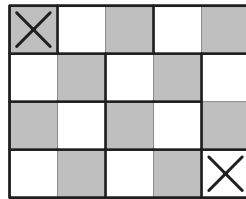
**Problem 2 — the checkerboard (the reason).** Counts (dark, light, covered?): A 4, 4, yes; B 3, 5, no; C 5, 5, yes; D 4, 6, no; E 6, 8, no; F 7, 7, yes. Rule: possible only when dark = light. Argument: neighbouring squares have different colours, so every domino covers exactly one dark and one light square;  $n$  dominoes cover  $n$  dark and  $n$  light; a board with unequal counts cannot be covered, whatever anybody tries. This explains all six boards in one line, including E. Push further: “Is ‘equal colours’ enough on its own?” No: cross out (1,1), (2,2), (3,4) and (4,3) on a 4-by-4 (6 dark and 6 light remain; pictured below) and the corner (4,4) is cut off with no neighbours. Equal colours is necessary; it is not always sufficient. Hints: (1) “Say the colours as you move the domino.” (2) “Count dark and light on the small pictures, not on the big board.”



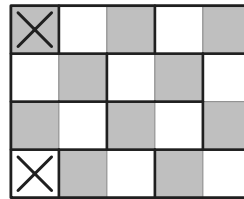
equal colours, but the corner (4,4) is cut off

**Problem 3 — predict, then test (4-by-5).** Counts and results: G (opposite corners (1,1), (4,5)): 9 dark, 9 light, prediction yes, result yes (23 covers, one pictured below). H (both top corners): 8 dark, 10 light, no, no. I (both left corners (1,1), (4,1)): 9 and 9, yes, yes (42 covers, one pictured below). Compare: “Same

board, opposite corners work this time; on the 4-by-4 they did not. Why?" On a board with an odd number of columns the two opposite corners have different colours ((1,1) and (4,5):  $1 + 1 = 2$  and  $4 + 5 = 9$ ); on the 4-by-4 they have the same colour (2 and 8). Hints: (1) "Predict from the counts only; then build." (2) For G and I: "Fill row by row with flat dominoes; a row with an odd number of free squares lends one square to the next row through an upright domino."



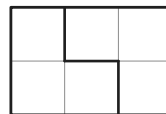
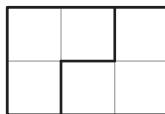
G



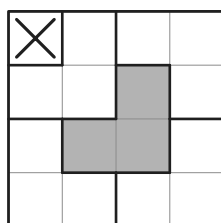
I

**Problem 4 — 6-by-6 and 8-by-8 minus opposite corners.** Answers: both impossible. 6-by-6: 16 dark, 18 light remain. 8-by-8: 30 dark, 32 light. Same reason as E, and the child should say it in the form "the same reason works on the bigger board": both corners are dark, so two more light squares than dark remain, and each domino takes one of each. For you: the full 8-by-8 has 12,988,816 domino covers, and removing any one dark and any one light square always leaves a coverable board (reserve 1 below). Hint: "Do not draw for long. Count."

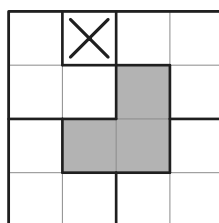
**Problem 5 — L-pieces on 2-by-3 and 3-by-3.** Answers: 2-by-3: exactly two covers (pictured below). 3-by-3: impossible. Argument: an L-piece always fits inside some 2-by-2 block; no 2-by-2 block of the 3-by-3 contains two corners of the board; so each L-piece covers at most one corner, three L-pieces cover at most three corners, and there are four. The area ( $9 = 3 + 3 + 3$ ) is fine; this "no" is a different kind of reason from the colour count, a "one corner per piece" count. Hints: (1) "Try to cover two corners with one L-piece." (2) "So how many corners can three L-pieces cover?"



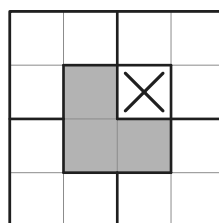
**Problem 6 — 4-by-4 minus one square, five L-pieces.** Answers: every one of the 16 squares works, and every one has exactly one cover; no second way exists. The covers for the corner (1,1), the edge square (1,2), the inside square (2,3) and another inside square, (3,3), are pictured below with the middle L-piece shaded. In every cover an L-piece sits in the middle four squares covering one square of each of the three quarters that do not contain the hole. Compare: "What do all your drawings have in common?" Hints: (1) "The quarter with the hole needs one L-piece filling its other three squares." (2) "Look at the middle four squares." (3) On the second-way search, after two minutes: "Where else could the middle L-piece go? Try each; something gets stuck."



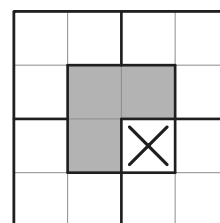
corner (1,1)



edge (1,2)



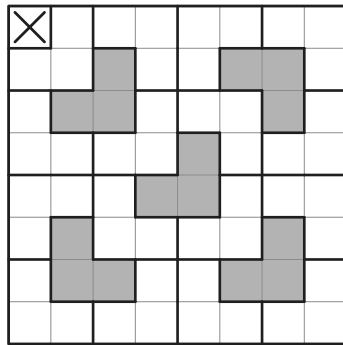
inside (2,3)



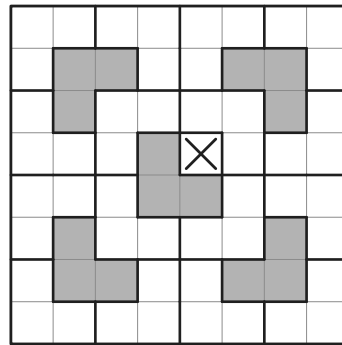
inside (3,3)

**Problem 7 — the centre trick.** Answer: it always works: after the middle L-piece, every quarter is a 2-by-2 with exactly one square missing, and one L-piece fills it. Since Problem 6 showed the cover is unique, this construction is the only cover. Ask: “Does the trick care which square the X is on?” (No; only which quarter.) “Would it work on a bigger board?” That is Problem 8. Hint: “Which three quarters do not have the X? Put one square of the middle L-piece in each.”

**Problem 8 — 8-by-8 with the centre trick.** Answer: it always works, with 21 L-pieces (63 squares); one L-piece in the middle, then the trick in each 4-by-4 quarter (one L-piece in the middle of each), then in each 2-by-2 block. Two examples are pictured below, with (1,1) crossed out and with (4,5) crossed out; the middle L-piece of the board and the middle L-pieces of the four quarters are shaded. Explanation form to ask for: “Why does it work for any crossed-out square?” Because the same reason works again on the bigger board: the middle L-piece turns one 8-by-8 problem into four 4-by-4 problems, and those are the ones already solved. A child may say “and then a 16-by-16 would work too”: yes, and 32-by-32, and so on (for you: this is Golomb’s induction; any  $2^n$ -by- $2^n$  board minus any square can be covered). Hints: (1) “First find the quarter with the X; the middle L-piece avoids it.” (2) “Inside each quarter, first find its own middle four squares.” (3) “Count the L-pieces at the end: the board has 63 free squares,  $63 = 3 \times 21$ .”

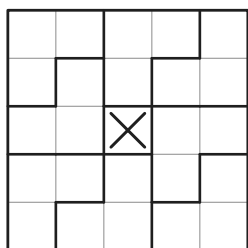


(1,1) crossed out

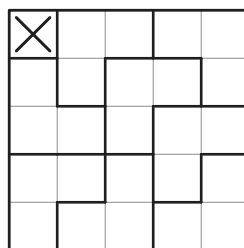


(4,5) crossed out

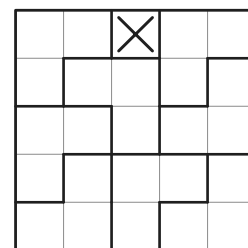
**Problem 9 — 5-by-5 minus one square, eight L-pieces.** Answers: centre: yes (32 covers). Corner (1,1): yes (8). Middle of the top edge (1,3): yes (16). Next to the corner (1,2): no. Example covers are pictured below. In general: exactly the nine squares with odd row and odd column work (the nine dotted squares); the other sixteen fail. Dotted board: an L-piece covers at most one dot (it fits inside a 2-by-2 block and every 2-by-2 block contains exactly one dot); eight L-pieces cover at most 8 dots; there are 9 dots; so the crossed-out square must be a dot. Compare with Problems 6 and 8: “2-by-2, 4-by-4, 8-by-8: every square works. 5-by-5: only nine of them. The pattern broke, and the dots say exactly where.” For you: on the 7-by-7 every square works again (checked for all 49 squares); among boards whose side is not a multiple of 3, the 5-by-5 is the only exception (Chu and Johnsonbaugh, 1986); when the side is a multiple of 3 the area rules everything out. Hints: (1) for the centre: “One L-piece in each corner first.” (2) For (1,2), after two tries: “Go to the dotted board now.” (3) “How many dots would eight L-pieces have to cover if every square except the X were covered?”



centre out



corner (1,1) out

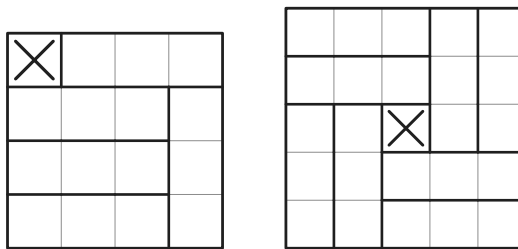


(1,3) out

**If the table finishes early (reserves, in this order).**

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2. Straight pieces (1-by-3), cut on demand. On the 4-by-4 board with one square crossed out, five straight pieces cover it only when the crossed square is a corner; on the 5-by-5 with eight straight pieces, only the centre works (both pictured below). Reason: colour the board in three colours along the diagonals (square  $(r, c)$  gets colour  $(r + c) \bmod 3$ ); a straight piece always covers one square of each colour; on the 4-by-4 the colours have 6, 5, 5 squares, so the hole must be in the 6-colour, and using the other diagonal direction as well, only the four corners are in both 6-colours; on the 5-by-5 the counts are 9, 8, 8 both ways and only the centre is in both 9-colours. Same board, three pieces, three different answers: dominoes (odd area, never), L-pieces (every square), straight pieces (corners only).



5-by-5, centre out

- 13

## 7. The share (about 6 minutes)

One child per table shows one thing; the adult at the table chooses in the last five minutes of table work.

- K–1: the shaded side-3 triangle with one blue on it (“white and grey, every time”) and the counts 6 and 3; or the 3-by-3 result grid with its tick pattern.
- Grades 2–3: the 3-by-3 result grid next to the shaded board (“the ticks are the dark squares”), then the 5-by-5 prediction grid.
- Grades 4–5: board E with the colour count (6 and 8), then the 8-by-8 covered by 21 L-pieces (“any square you cross out”).

Closing line for the organizer: “Every ‘no’ today came with a reason you could check: an odd number of squares; two colours that did not match; corners or dots that no piece can share. Mathematicians call these invariants, and finding a new one is how you prove that something can never be done.” Take-home questions (say them, do not hand them out): for K–1, “can you cover the star with reds only?” (no); for 2–3, “the 2-by-5 board: how many different ways with dominoes?” (8); for 4–5, “which squares of a 7-by-7 board can be the hole for L-pieces?” (all of them).

## 8. The mathematics behind the session (for the two mathematicians)

**Invariants.** Every “no” in the packet is a counting invariant. The parity of the area (3-by-3, side-3 triangle) is the crudest. The checkerboard argument is the bipartition of the grid graph: a domino tiling is a perfect matching, so the two colour classes must be equal; on the triangular lattice the up/down orientation of unit triangles is the same bipartition (adjacent unit triangles always point opposite ways), which is why the K–1 blue-rhombus problems are the pattern-block form of the mutilated checkerboard. The L-tromino “no”s (3-by-3, and the 16 bad holes of the 5-by-5) are a different invariant: a set of cells no piece can cover twice (the four corners; the nine cells with both coordinates odd), so  $k$  pieces cover at most  $k$  of them. The straight-tromino reserve uses Golomb’s three-colouring along diagonals. Colour arguments are linear invariants; they are necessary conditions and often not sufficient (equal colours but an isolated corner; the star with 6 white and 6 grey units that red trapezoids still cannot tile, because each of its six points needs its own trapezoid). Where colourings fail there are stronger tools: Conway and Lagarias’s tiling groups (1990) and Thurston’s height functions (Amer. Math. Monthly 1990), which for instance prove which triangular arrays of hexagons can be tiled by three-in-a-row “tribones”, and show that the domino and lozenge tilings of a simply connected region are connected under flips (checked by search for the rectangles in this packet and for the side-2 hexagon: no tiling of any of them is flip-free).

**Matchings and counting.** Domino tilings of a region are perfect matchings of its grid graph, so tileability is decidable in polynomial time (Hall’s theorem gives the exact condition; the colour count is the case where the deficient set is a whole colour class), and Gomory’s loop argument (reserve 1 at the 4–5 table) is a Hamiltonian-cycle proof that on the 8-by-8 any dark-light pair can be removed. Counting is Kasteleyn’s and Temperley–Fisher’s Pfaffian method: the 8-by-8 board has 12,988,816 domino tilings; 2-by- $n$  boards give the Fibonacci numbers; 3-by- $n$  boards by straight trominoes satisfy  $a(n) = a(n-1) + a(n-3)$ . Lozenge tilings of the side- $n$  hexagon by blue rhombi are plane partitions in an  $n \times n \times n$  box (MacMahon’s product formula); the side-2 hexagon has 20, one for each stack of 0 to 8 cubes, distributed 1, 1, 3, 3, 4, 3, 3, 1, 1 by cube count, and every tiling uses 4 rhombi of each of the three orientations. For large regions a random tiling shows the arctic-circle phenomenon (the Aztec diamond; Jockusch, Propp and Shor; Cohn, Kenyon and Propp), which leads straight into current research on random tilings and dimer models.

**L-trominoes.** Golomb (1954) proved by the centre-trick induction that every  $2^n$ -by- $2^n$  board minus one square is L-tileable; Chu and Johnsonbaugh (1986) settled all deficient  $n$ -by- $n$  boards: for  $n$  not divisible by 3 and  $n \neq 5$ , every hole works (checked here for  $n = 2, 4, 7, 8$ ); for  $n = 5$  exactly the nine holes with both coordinates odd; for  $n$  divisible by 3 the area  $n^2 - 1$  is not a multiple of 3. The uniqueness of the tiling of the 4-by-4 minus one is special to that size (the 5-by-5 minus the centre already has 32 tilings, the 8-by-8 minus a corner has 30,355).

**Hardness and where it leads.** For dominoes the tiling question is a matching problem, but for most other tile sets deciding whether a finite region (with holes allowed) can be tiled is NP-complete, already for tile sets consisting only of trominoes and the 2-by-2 square (Moore and Robson 2001; Pak and Yang 2013 for simply connected regions and rectangular tiles). Tiling the whole plane is undecidable (Berger 1966), which is the origin of aperiodic tile sets, from Penrose's to the single aperiodic "hat" tile found in 2023. Good next reading: Ardila and Stanley, "Tilings" (Math. Intelligencer 2010); Golomb, *Polyominoes*; Thurston, "Conway's tiling groups".

**What to say if a child asks "is there a shape nobody knows how to tile?"** Yes: for many regions nobody knows a quick way to decide, and that is a theorem, not ignorance.